# MAT1011 Honors Calculus I

# Homework 5

2026 Fall

Show your work. Use only methods and results covered in class. All definite integrals are Riemann integrals. The Fundamental Theorem of Calculus may not be used. Throughout, $\mathbb{N} = \{1, 2, 3, \ldots\}$ and all angles are in radians.

## 1. Compulsory Exercises

**Question 1.** Let

$$f(x) = \frac{x}{1+x^2}, \qquad x \in \mathbb{R}.$$

(a) Determine where $f$ is strictly increasing or strictly decreasing, and find all local maximum and minimum points and their values.

(b) Determine where $f$ is strictly convex or strictly concave, and find the coordinates of all points of inflection.

**Question 2.** Find all antiderivatives of each function on the stated interval.

(a) $f(x) = \dfrac{2x+1}{(x^2+x+1)^{3/2}}, \qquad x \in \mathbb{R}.$

(b) $g(x) = \sin^3 x \cos^2 x, \qquad x \in \mathbb{R}.$

(c) $h(x) = \tan^4 x, \qquad -\pi/2 < x < \pi/2.$

(d) $u(x) = |x^2 - 1|, \qquad x \in \mathbb{R}.$

**Question 3.** For each $n \in \mathbb{N}$, let $P_n = \{x_0, \ldots, x_n\}$ be the partition of $[0, 1]$ defined by

$$x_k = \frac{k^2}{n^2}, \qquad k = 0, 1, \ldots, n.$$

(a) Find $\|P_n\|$ and compute the Riemann sum

$$R_n = \sum_{k=1}^{n} \sqrt{x_k}\,(x_k - x_{k-1}).$$

(b) Evaluate $\displaystyle\int_0^1 \sqrt{x}\,dx$ using these sums. Explain why the limit of these sums equals the integral.

**Question 4.**

(a) Let $f : [a, b] \to \mathbb{R}$ be nondecreasing, where $a < b$. For a partition $P$ of $[a, b]$, let $S(P)$ and $s(P)$ be the upper and lower sums of $f$. Prove that

$$S(P) - s(P) \le \big(f(b) - f(a)\big)\|P\|.$$

Deduce that $f$ is Riemann integrable on $[a, b]$.

(b) Evaluate

$$\int_0^2 [x^2]\,dx,$$

where $[t]$ denotes the greatest integer not exceeding $t$.

**Question 5.** Determine whether each function is uniformly continuous on its stated domain. Prove each conclusion.

(a) $f(x) = \sqrt{x}$ on $[0, \infty)$.

(b) $g(x) = \sin(x^2)$ on $\mathbb{R}$.

**Question 6.** Let $f : [a, b] \to \mathbb{R}$ be bounded, where $a < b$.

(a) If $f$ is Riemann integrable, prove that $|f|$ and $f^2$ are Riemann integrable.

(b) If $|f|$ and $f^2$ are both Riemann integrable, must $f$ be Riemann integrable? Give a proof or a counterexample.

**Question 7.** Define $w : [0, 1] \to \mathbb{R}$ by $w(0) = 0$ and

$$w(x) = (-1)^n \quad \text{if } 2^{-(n+1)} < x \leq 2^{-n}, \qquad n = 0, 1, 2, \ldots.$$

For a partition $P$ of $[0, 1]$, let $S(P)$ and $s(P)$ be the upper and lower sums of $w$.

(a) Prove that, for every partition $P$ and every integer $N \geq 1$,

$$S(P) - s(P) \leq 2^{1-N} + 4N\|P\|.$$

Deduce that $w$ is Riemann integrable on $[0, 1]$.

(b) Compute $\displaystyle\int_0^1 w(x)\, dx$.

**Question 8.** Let $f : \mathbb{R} \to \mathbb{R}$ be differentiable. Suppose that there is a constant $m > 0$ such that $f'(x) \geq m$ for every $x \in \mathbb{R}$.

(a) Prove that, for every $t \in \mathbb{R}$, the equation $f(x) = t$ has exactly one real solution.

(b) For each $t \in \mathbb{R}$, let $g(t)$ denote the unique solution of $f(x) = t$. Prove that

$$|g(s) - g(t)| \leq \frac{1}{m}|s - t| \qquad (s, t \in \mathbb{R}).$$

Deduce that $g$ is uniformly continuous on $\mathbb{R}$. Prove also that $g$ is differentiable on $\mathbb{R}$ and that

$$g'(t) = \frac{1}{f'(g(t))} \qquad (t \in \mathbb{R}).$$

## 2. Extra Exercises

**Question 9.** Let $u, v : [a, b] \to \mathbb{R}$ be continuous, where $a < b$, and define

$$h(x) = \begin{cases} u(x), & x \in \mathbb{Q}, \\ v(x), & x \notin \mathbb{Q}. \end{cases}$$

Prove that $h$ is Riemann integrable on $[a, b]$ if and only if $u(x) = v(x)$ for every $x \in [a, b]$.

**Question 10.** Let $f : (0, 1) \to \mathbb{R}$ be uniformly continuous. Prove that there is exactly one continuous function $F : [0, 1] \to \mathbb{R}$ such that

$$F(x) = f(x) \qquad (0 < x < 1).$$

**Question 11.** Let $f : [a, b] \to \mathbb{R}$ be Riemann integrable, where $a < b$, and suppose $|f(x)| \leq M$ on $[a, b]$ for some $M > 0$. Let $\phi : [-M, M] \to \mathbb{R}$ be continuous. Prove that the function $x \mapsto \phi(f(x))$ is Riemann integrable on $[a, b]$.

## 3. Thomas Textbook Exercises

**Newton's method.** Exercises 4.6: 8, 24.

**Antiderivatives.** Exercises 4.7: 56, 68, 88, 90.